

Solution Architecture

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of Green-Predict-Pro across Presentation, Application and Data layers.

Presentation / Client Layer — Web App (HTML, CSS, Bootstrap, JavaScript) accessible on desktop and mobile browsers

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API Gateway / Web Server — Flask development server (production: Gunicorn behind Nginx)

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Auth Service (Flask-Login, hashed passwords) | Core Logic Service (Data preprocessing + Random Forest ML model) | External API Integrations (optional Weather API)

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Data / Storage Layer — SQLite / MySQL user & history DB, Crop Recommendation CSV, model.pkl & scaler.pkl artifacts

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects farmer inputs and displays crop recommendation and insights.	HTML5, CSS3, Bootstrap, JavaScript, Jinja2
API / Web Server	Routes user requests, handles session and forwards data to the ML pipeline.	Flask, Gunicorn, Nginx
Core Logic / ML Service	Validates inputs, preprocesses features and runs the trained classifier to produce recommendations.	Python, scikit-learn, NumPy, Pandas
Database	Stores users, prediction history and metadata.	SQLite (dev) / MySQL (prod)

Model Store	Serialized model and scaler artifacts loaded at server start.	joblib / pickle files
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